Final technical audit

Review date: 2 November 2026. This is a research preparation audit, not an IEEE decision. The repository and local publication source were inspected; original model artifacts and authenticated training-source snapshots remain unchanged.

A01 Early detection title

Severity: Critical. Title implies a verified prediction cutoff; no predictor timestamps exist.

Prior revision evidence: Renamed retrospective classification; revised abstract, objectives and conclusion.

Remaining scope or author action: Needs new timestamped cohort for prospective claim.

A02 Absences availability

Severity: High. Aggregate absences may span the outcome period.

Prior revision evidence: Retuned no-absences models under the same nested folds.

Remaining scope or author action: Other feature timestamps remain unknown.

A03 Small holdout

Severity: High. Only 26 positive outcomes produce wide rate uncertainty.

Prior revision evidence: Wilson recall intervals and conditional paired bootstrap intervals added.

Remaining scope or author action: Larger independent cohorts required.

A04 Tuning optimism

Severity: High. Five-fold maximum scores reuse the folds that choose settings.

Prior revision evidence: Added 5-fold x 2-repeat outer validation with complete inner search on original training rows.

Remaining scope or author action: Dependent folds are not independent cohorts.

A05 Known holdout

Severity: High. Revision was designed after original test results were known.

Prior revision evidence: Extensions labeled exploratory; fixed original row IDs, no threshold/default replacement.

Remaining scope or author action: Fresh validation remains necessary.

A06 Objective baseline

Severity: High. Always-At-Risk exceeds both original models on test F2.

Prior revision evidence: Retained baseline; explicitly interpret precision, recall and workload.

Remaining scope or author action: No institution-specific cost or capacity elicitation.

A07 Historical rule baseline

Severity: Medium. Original study lacked a past-failure rule.

Prior revision evidence: Added failures > 0 rule without tuned cutoff.

Remaining scope or author action: Timing of recorded history still requires confirmation.

A08 Probability benchmark

Severity: Medium. Hard constant baselines are weak probability comparators.

Prior revision evidence: Added training prevalence constant for Brier comparison.

Remaining scope or author action: Hard heuristic scores are not probabilities.

A09 Model diversity

Severity: Medium. Two models alone do not test nonlinear ensemble behavior.

Prior revision evidence: Added bounded RF and GB searches using same partitions and scoring.

Remaining scope or author action: Search budgets differ; no universal family ranking.

A10 No calibration evidence

Severity: High. Brier and displayed percentages do not establish calibration.

Prior revision evidence: Added nested-search sigmoid calibration ensembles and fixed-bin reliability diagnostics.

Remaining scope or author action: Small test bins; ensemble effect confounds pure recalibration effect.

A11 Subgroup fairness

Severity: High. No sex, address or school error assessment existed.

Prior revision evidence: Added denominators, recall and FPR with Wilson intervals.

Remaining scope or author action: No fairness certification, causal analysis or mitigation.

A12 Sensitive predictors

Severity: Medium. Private, health and proxy inputs create use concerns.

Prior revision evidence: Retuned reduced 15-feature sensitivity scenario.

Remaining scope or author action: Removal does not eliminate proxies or establish fairness.

A13 Prior-failure dependence

Severity: Medium. Unclear whether other information adds predictive value.

Prior revision evidence: Added no-failures ablation and simple-rule comparison.

Remaining scope or author action: No causal effect inferred.

A14 Threshold selection

Severity: Medium. F2 search at 0.5 may hide workload trade-offs.

Prior revision evidence: Added descriptive 0.1-0.9 test sensitivity curves.

Remaining scope or author action: No test-based threshold tuning; capacity study pending.

A15 Paired uncertainty

Severity: Medium. Ranking differences had point estimates only.

Prior revision evidence: 2,000 paired stratified bootstraps; exact LR-DT McNemar test.

Remaining scope or author action: Intervals condition on fixed models/prevalence, omit cluster/retraining uncertainty.

A16 Fold independence

Severity: Medium. Repeated folds overlap in fit rows and students.

Prior revision evidence: Means and sample SD only; no naive fold-based significance test.

Remaining scope or author action: External replication remains needed.

A17 Importance bias

Severity: Medium. Training impurity can favor split-rich inputs.

Prior revision evidence: Added raw-feature AP permutation on outer validation folds.

Remaining scope or author action: Correlation and out-of-distribution permutations remain limitations.

A18 Explanation stability

Severity: Medium. One fitted coefficient/tree ranking is unstable on small data.

Prior revision evidence: Saved per-fold LR coefficients and validation permutation values.

Remaining scope or author action: No causal or intervention interpretation.

A19 Odds ratio units

Severity: Medium. Numeric and categorical coefficients have different units.

Prior revision evidence: Clarified per-training-SD effects and omitted categorical references.

Remaining scope or author action: No conventional unpenalized inferential CI claimed.

A20 Ordinal codes

Severity: Medium. LR treats ordinal steps as equally spaced.

Prior revision evidence: Preserved actual encoding and stated assumption.

Remaining scope or author action: Alternative categorical/spline coding is a possible future study.

A21 External validation

Severity: High. No independent school/cohort evaluation.

Prior revision evidence: Scope limited; public future benchmark suitability documented.

Remaining scope or author action: Requires compatible independent data; not repairable with random resampling.

A22 Temporal validation

Severity: High. Random split does not test future cohorts.

Prior revision evidence: Explained cutoff and cohort validation requirements; assessed OULAD compatibility.

Remaining scope or author action: New temporal data needed; OULAD is a separate task.

A23 School clustering

Severity: Medium. Shared school/class context may overstate independence.

Prior revision evidence: Reported assumptions; school subgroup counts retained.

Remaining scope or author action: Only two schools and no class IDs; no cluster-robust inference claimed.

A24 Zero grades

Severity: Medium. 38 zero grades may mix absence/withdrawal/achievement.

Prior revision evidence: Retained original target and explicitly reported ambiguity.

Remaining scope or author action: Administrative meaning needs source confirmation.

A25 Dataset identity

Severity: Medium. UCI summary has 649 language records versus 395 mathematics rows.

Prior revision evidence: Verified actual shape, hash, labels, missingness and full-row duplicates.

Remaining scope or author action: No mathematics/language pooling or false external validation.

A26 Novelty

Severity: Medium. Two standard algorithms and an app offer limited algorithmic contribution.

Prior revision evidence: Reframed as reproducibility and sensitivity audit with bounded claims.

Remaining scope or author action: Venue fit and sufficient significance require editorial assessment.

A27 Literature currency

Severity: Medium. Original bibliography mostly predates 2023.

Prior revision evidence: Added six relevant 2023-2026 works; DOI and official-source records saved.

Remaining scope or author action: Focused update, not a systematic literature review.

A28 Environment claim

Severity: Medium. Paper says Python 3.12.14 while current environment is 3.12.10.

Prior revision evidence: Reported observed interpreter and retained historical metadata separately.

Remaining scope or author action: Historical run environment not retroactively altered.

A29 Prototype privacy

Severity: Medium. Current interface requests a name unnecessary for inference.

Prior revision evidence: Inspected session/code behavior; recommended data minimization.

Remaining scope or author action: App is not modified or represented as validated deployment.

A30 Unsupported declarations

Severity: High. Funding/conflicts/ethics/repository placeholders were unresolved.

Prior revision evidence: Removed placeholders and invented acknowledgment; tracked author actions separately.

Remaining scope or author action: Authors must provide actual declarations and archive URL.

A31 Authorship and biographies

Severity: High. Author identity must remain and biographies are required by Access.

Prior revision evidence: Names/order/affiliation/correspondence preserved; no biographies invented.

Remaining scope or author action: All authors must supply approved biographies and verify identity details.

A32 AI disclosure

Severity: Medium. Original AI declaration was a placeholder.

Prior revision evidence: Factual acknowledgment describes AI assistance and cites the system.

Remaining scope or author action: Authors must approve disclosure and verify all assisted material.

A33 Formatting and equations

Severity: Medium. Original PDF had large blank areas and was not the Access template.

Prior revision evidence: Built editable DOCX from official template; added native equations, numbered figures/tables.

Remaining scope or author action: Final Word/PDF layout and content checked; submission still awaits author actions.

A34 Unknown categories

Severity: Low. Ignored one-hot levels can encode like a reference category.

Prior revision evidence: Documented pipeline behavior and inspected application checks.

Remaining scope or author action: Unknown cohort categories and range drift require deployment validation.

Additional repository findings and corrections

F01 corrected: README and current report generator still used Early Detection. Current README, report titles and app description now use retrospective framing; historical setup and training snapshots retain provenance.

F02 corrected: manuscript said no public code was available although the repository is public. Final availability statement links the repository and distinguishes an unpublished release from a DOI archive.

F03 corrected: original verifier recomputed only model nested predictions and checked subgroup totals, not all baseline fold calculations. audit_final.py independently recomputes all 140 fold records including 40 baseline records, summary sample SD, exact subgroup confusion matrices and Wilson intervals, saved-model probabilities, calibration-bin counts and ensemble membership.

F04 corrected: full experiment script wrote into existing publication/results. An explicit RISK_PUBLICATION_OUTPUT override allows isolated reruns without changing the protocol or historical outputs.

F05 corrected: no citation metadata, release notes or licence-status document. Added CITATION.cff, rights notice, proposed release notes and Zenodo steps. No new reuse licence was imposed without rights-holder approval.

F06 preserved: duplicate project/reproducibility copies and historical papers are intentional evidence. Local duplicate ZIPs, environments, caches, review renders and credentials remain ignored. Useful source/model artifacts are retained.

F07 residual: stored historical XML/log/environment files contain old local paths. They are diagnostic provenance, not portable setup instructions. New documentation uses relative paths; no credentials or private identifiable student dataset were found in the targeted token scan. UCI anonymous records remain attributed.

F08 corrected: final document builder accepts a separate output stem, retaining Revised_Manuscript exports. Word is used for the final matching PDF; no LaTeX manuscript source exists, so no synthetic TEX conversion is represented as original source.

F09 residual: author biographies, funding/conflict declarations, ethics determination, ORCID details, licence and correspondence confirmation require human evidence. The final manuscript deliberately contains no invented declarations; it remains author-action pending.

F10 corrected: final PDF authoring needs ReportLab, absent from the prior authoring dependency list. Added reportlab==4.4.9 and a separate jsonschema validation requirement without changing the numerical project lock. The new reference verifier uses the standard library; the earlier requests-based verifier is historical and is not the documented current command.

F11 residual: live registry queries retrieved all DOI-backed references with matching titles and recorded full author/date/journal metadata. Direct retrieval of the 2008 paper host failed certificate validation, and the AI-system page rejected the scripted request. UCI independently confirms the original paper authors/title/year; the AI-system page was reachable through the browser retrieval service. These fallbacks do not verify the original paper page range or prove absence of retractions. See final/Reference_Review.md.

Verification evidence

See final/experiment_verification.json for executed independent checks and fresh-run comparisons, final/reference_verification.json for live bibliography metadata, and final/document_verification.json for final DOCX/PDF parity. Original 27 tests and Streamlit startup are rerun for this revision. Failed or incomplete checks must be reported rather than converted into success claims.

Remaining rejection risks

Limited novelty from standard algorithms on a heavily used dataset; small/cohort-specific evidence; inability to establish predictor timing or independent transportability; dependent resampling and post-hoc exploratory extensions; incomplete author declarations and biographies. No amount of local testing repairs the unavailable external cohort or timestamps.